

Tharun Kumar Bettadalli Girish

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EDUCATION

The University of Texas at Arlington

Master of Science in Data Science | GPA: 3.6

Texas, USA

Jan 2024 - Dec 2025

Cambridge Institute of Technology, VTU

Bachelor of Engineering in Computer Science | CGPA: 8.0

Bengaluru, India

Aug 2019 - May 2023

EXPERIENCE

University of Texas at Arlington – SEAR Lab

Texas, USA

Research Assistant (Data Analytics)

Jan 2026 – Present

- Analyzed 15,000+ RFID commissioning records across multiple deployment environments, quantifying 12% variance in read accuracy and isolating operational performance gaps.
- Built and automated Python and SQL data validation workflows to clean and structure 50+ experimental datasets, reducing preprocessing time by 40% and improving downstream data reliability.
- Produced statistical performance analyses and interactive dashboards supporting 3 concurrent research initiatives, enabling 8–15% optimization improvements in controlled system evaluations.

Samsung SEED Pvt Ltd

Bengaluru, India

Data Annotator

May 2023 - Dec 2023

- Executed semantic segmentation on millions of image pixels, classifying complex scenes with advanced interaction techniques that reduced annotation error rates by 20% and improved downstream model accuracy by 15%.
- Leveraged Samsung R&D's proprietary annotation tool Datagen to refine image annotation accuracy by 30%, enhancing deep learning model training efficiency by 25% across multiple AI initiatives serving millions of users.
- Optimized annotation workflows and data processing pipelines across cross-functional AI teams, reducing time-to-production for computer vision models and streamlining operations for production deployment at scale.

Varcons Technology Pvt Ltd

Bengaluru, India

Data Science Intern

Apr 2022 - Sep 2022

- Designed and implemented an end-to-end recommendation pipeline on 500K+ user records, delivering personalized diet insights to 10K+ users and increasing engagement by 15% through behavioral pattern analysis.
- Developed SQL-based ETL data pipelines using MySQL and PostgreSQL for large-scale user data, reducing query latency by 25% and enabling real-time analytics for ML applications.
- Created backend analytics services using Python and Flask to support scalable data processing, reporting, and KPI tracking, partnering with product and engineering teams to translate business requirements into measurable data science solutions.

PROJECTS

GenAI-Powered Academic Analytics and Semantic Question-Answering System (RAG)

- Implemented a production-ready GenAI Q&A system integrating SQL data, FAISS semantic search, and LLMs, implementing intent-aware query routing to direct factual queries to SQL retrieval and interpretive queries to RAG.
- Achieved 85% Top-5 retrieval accuracy on semantic search benchmarks and reduced information discovery time by 40% through optimized vector embeddings and recommendation logic, deployed system using Gradio and PostgreSQL.

E-Commerce Customer Analytics & Retention Analysis | SQL, Python, Tableau

- Built an end-to-end e-commerce analytics project using SQL, Python, and Tableau to analyze ~100K+ transactions, uncovering revenue trends, customer behavior, and key drivers across products and regions.
- Performed cohort analysis and RFM segmentation to identify extremely low retention (<1%) and high dependence on at-risk customers, delivering actionable insights to improve customer retention and long-term revenue growth.

Real-Time Warehouse Inventory and AI-Assisted Logistics Optimization System

- Designed and engineered a real-time inventory tracking system in Python and Spark across 50+ warehouses, achieving sub-second latency and 100% sync accuracy, and developed ML-based incident resolution models reducing resolution time by 21%.
- Improved warehouse-to-delivery workflows using routing analytics, reducing logistics costs by 7.5% and improving pipeline reliability with data quality monitoring.

SKILLS

Programming and Query: Python, SQL, PySpark, R

Data Engineering: Spark, Databricks, HDFS, ETL, Snowflake, DynamoDB, PostgreSQL, MySQL

Machine Learning & GenAI: Scikit-learn, XGBoost, PyTorch, TensorFlow, FAISS, RAG, LLMs

Analytics & Tools: Tableau, Power BI, Advanced Excel, Matplotlib, Seaborn, Plotly, GitHub, Docker, Jenkins

CERTIFICATIONS

Data Analytics and Databases on AWS | Microsoft Certified: Azure Fundamentals (AZ-900) | IBM Excel for Data Analysis

LEADERSHIP AND INVOLVEMENT

Co-Founder, DATA (Data Analytics and Technology Association), UTA Data Science Department

Sep 2024 - Dec 2025

- Led datathons and analytics workshops for 200+ participants, delivering 15+ data science projects and hands-on training.
- Scaled a 150+ member analytics community at UTA, coordinating cross-functional teams, industry mentorship initiatives.